AN IMPROVED LOWER BOUND FOR SETS WITH NO CUBE DIFFERENCES

ANONYMOUS AUTHORS

Abstract. Let $D_3(N)$ be the largest size of a subset of $\{1,\dots,N\}$ no two of whose elements differ by a nonzero cube. Ruzsa's construction gives $D_3(N) \gg N^{\gamma(m)}$ with $\gamma(m) = \frac{1}{3}\bigl(2+\log \alpha_3(m)/\log m\bigr)$ for every squarefree m, where $\alpha_3(m)$ is the largest size of a subset of $\mathbb{Z}/m\mathbb{Z}$ with no nonzero cube difference. The best exponent previously recorded, $\gamma(91) = \frac{1}{3}(2+\log 14/\log 91) = 0.861681\ldots$, comes from a 14-element set modulo $91 = 7\cdot 13$ found by Lewko. We exhibit an explicit 116-element set modulo $3367 = 7\cdot 13\cdot 37$ with no nonzero cube difference, which gives

$$D_3(N) \gg N^{\gamma}, \qquad \gamma = \tfrac{1}{3}\Bigl(2+\frac{\log 116}{\log 3367}\Bigr) = 0.861763\ldots.$$

The modulus was located by a structural analysis of the modular problem: only primes $\equiv 1 \pmod 3$ can occur, the "base m^3" variant of the construction gives nothing new, and a strong-product ceiling $\alpha_3(7m') \le \lfloor 7\alpha_3(m')/2\rfloor$ together with exact values of $\alpha_3(p)$ for all primes $p \le 163$ excludes every modulus $7p$ with $p \le 163$ and singles out 3367 as the natural candidate. The certificate was found by a tabu search over the fibres of the decomposition $\mathcal{P}_3(\mathbb{Z}/3367\mathbb{Z}) \cong C_7 \boxtimes \mathcal{P}_3(\mathbb{Z}/481\mathbb{Z})$; every finite claim is verified by an accompanying standard-library script.

1. Introduction

A theorem of Sárközy [10] (see also Furstenberg [3] for $k=2$) states that if $A \subseteq [N] := \{1,\dots,N\}$ contains no two distinct elements whose difference is a k-th power, then $|A| = o(N)$. The strongest quantitative upper bounds for general k are due to Balog, Pelikán, Pintz and Szemerédi [1]; for $k=2$ the recent work of Green and Sawhney [4] gives $|A| \ll N\exp(-c\sqrt{\log N})$.

In the other direction, the only constructions of polynomial size known are digital constructions going back to Ruzsa [9]. Fix $k \ge 2$ and a squarefree $m \ge 2$, and let $R \subseteq \mathbb{Z}/m\mathbb{Z}$ be a set no two of whose distinct elements differ by a k-th power modulo m. Ruzsa proved that there is a set $A \subseteq [N]$ with no nonzero k-th power difference and

$$|A| \gg_m N^{\gamma}, \qquad \gamma = \frac{k-1+\log_m |R|}{k}. \tag{1}$$

For squares, the choice $m=65$, $|R|=7$ is due to Ruzsa, and the choice $m=205$, $|R|=12$, giving $\gamma = 0.7334\ldots$, is due to Beigel–Gasarch [2] and Lewko [6]. For cubes, the exponent $0.8616\ldots$ is recorded in Younis [11, §1] as the result of a calculation of Lewko [6]; it coincides with the value of (1) for $k=3$, $m=91$, $|R|=14$, namely

$$\gamma_{91} := \frac{1}{3}\Bigl(2+\frac{\log 14}{\log 91}\Bigr) = 0.861681779699457\ldots, \tag{2}$$

and we verify below (exactly) that 14 is the largest size of a cube-difference-free set modulo 91. Younis [11] extended Ruzsa's construction to polynomial differences and to the nonlinear Roth configuration; Naslund [8] adapted it to function fields and studied the independence numbers of the relevant generalised Paley graphs.

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Write $D_3(N)$ for the largest size of a subset of $[N]$ with no two elements differing by a nonzero cube, and

$$\alpha_3(m) := \max\bigl\{|R| : R \subseteq \mathbb{Z}/m\mathbb{Z},\ (R-R)\cap\{x^3 : x\in\mathbb{Z}/m\mathbb{Z}\} = \{0\}\bigr\},$$

so that (1) reads $D_3(N) \gg_m N^{\gamma(m)}$ with $\gamma(m) = \frac{1}{3}\bigl(2 + \log\alpha_3(m)/\log m\bigr)$. Our main result is:

1.1. **Theorem.** *$\alpha_3(3367) \geq 116$: the 116-element set $R_{3367} \subseteq \mathbb{Z}/3367\mathbb{Z}$ listed in Section 6 has no two elements differing by a nonzero cube modulo $3367 = 7\cdot 13\cdot 37$. Consequently*

$$D_3(N) \ \gg\ N^{\gamma}, \qquad \gamma = \gamma(3367) = \frac{1}{3}\Bigl(2 + \frac{\log 116}{\log 3367}\Bigr) = 0.861763131630587\ldots,$$

which improves $\gamma_{91} = 0.861681779\ldots$ by $8.1\cdot 10^{-5}$.

The improvement is small, as is to be expected: by a result of Matolcsi and Ruzsa [7] (see [8, §2.2]), $\alpha_3(m) \ll_\varepsilon m^{1/2+\varepsilon}$ for squarefree m, so that $\limsup \gamma(m) \leq 5/6$ and any improvement inside Ruzsa's framework must come from a small modulus, where the gains come from the superadditivity of independence numbers under strong products. The point of this note is that such a modulus exists and can be identified systematically. Put

$$r := \frac{\log 14}{\log 91} = 0.585045339098370\ldots; \tag{3}$$

then $\gamma(m) > \gamma_{91}$ if and only if $\alpha_3(m) > m^r$, and we let $T(m)$ denote the least integer exceeding m^r (so $T(91) = 15$ and $T(3367) = 116$, as $3367^r = 115.770\ldots$).

Structure of the modular problem. Let $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$ be the graph on $\mathbb{Z}/m\mathbb{Z}$ in which $x \sim y$ when $x - y$ is a nonzero cube modulo m (this is symmetric since $-1 = (-1)^3$), so that $\alpha_3(m)$ is its independence number. By the Chinese remainder theorem $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$ is the strong product of the graphs $\mathcal{P}_3(\mathbb{F}_p)$, $p \mid m$ (Lemma 3.1). We prove:

1.2. **Theorem.**
(i) *If m is squarefree and some prime $p \mid m$ satisfies $p \not\equiv 1 \pmod 3$, then $\alpha_3(m) = \alpha_3(m/p)$; consequently $\gamma(m) \leq \gamma(m/p)$, with equality only if $\alpha_3(m/p) = 1$.*
(ii) *If m is squarefree and $3 \nmid m$, then $\alpha_3(m^3) = m^2\alpha_3(m)$. In particular the "base m^3" form of the construction (Younis* [11, Theorem 1.7] *with $F(x) = x^3$) gives exactly the exponent $\gamma(m)$.*
(iii) *(Ceiling.) If $m = m_1m_2$ with $\gcd(m_1, m_2) = 1$ and $\omega_3(m_1)$ denotes the clique number of $\mathcal{P}_3(\mathbb{Z}/m_1\mathbb{Z})$, then*

$$\alpha_3(m) \leq \Bigl\lfloor \frac{m_1}{\omega_3(m_1)}\,\alpha_3(m_2)\Bigr\rfloor.$$

In particular $\alpha_3(7m') \leq \lfloor 7\alpha_3(m')/2\rfloor$ for every m' coprime to 7.

Exact data and the choice of the modulus. We computed $\alpha_3(p)$ and $\omega_3(p)$ exactly for every prime $p \equiv 1 \pmod 3$ with $p \leq 163$ (Table 1), and $\alpha_3(91) = 14$ exactly. Feeding these values into Theorem 1.2(iii) gives:

1.3. **Theorem.** *Let m be squarefree.*
(i) *If $7 \mid m$ and $\gamma(m) > \gamma_{91}$, then $m' = m/7$ satisfies $\alpha_3(m') > c\,(m')^r$ with $c = 2\cdot 7^{r-1} = 0.891974\ldots$; that is, $\alpha_3(m')$ must itself come within 11% of $(m')^r$.*
(ii) *No modulus $m = 7p$ with $p \leq 163$ prime satisfies $\gamma(m) > \gamma_{91}$.*
(iii) *Among the 153 moduli pq with $p < q \leq 163$ primes $\equiv 1 \pmod 3$, the ceiling of Theorem 1.2(iii) alone excludes all but the 38 listed in Table 2, none of which is divisible by 7.*
(iv) *$\alpha_3(3367) \leq 157$ unconditionally, and $\alpha_3(3367) \leq 119$ if $\alpha_3(481) = 34$.*

Theorem 1.3(i) explains why 3367 is the natural candidate: the cofactor $481 = 13\cdot 37$ is, among all cofactors we examined, the only one satisfying $\alpha_3(m') > 0.892\,(m')^r$ (here $0.892\cdot 481^r = 33.1$ and the searched value is $\alpha_3(481) \geq 34$), and the ceiling $\lfloor 7\cdot 34/2\rfloor = 119$ leaves room above $T(3367) = 116$. Section 5 describes the search that found the certificate; Section 6 lists it.

We are careful to separate what is proved from what is searched. Theorems 1.1, 1.2 and 1.3 are proved, the finite verifications being reproduced by the scripts described in Appendix A. The values reported in Section 5 for moduli other than those covered by exact computation are lower bounds produced by local search, and we claim no optimality for them; in particular $\alpha_3(481) = 34$ and $\alpha_3(3367) = 116$ are not claimed.

2. Ruzsa's construction

We include a complete proof of (1) for $k = 3$, following Ruzsa [9] in the form given by Younis [11, Theorem 1.6], since the argument determines exactly which modular quantity is being optimised. Throughout, $m \geq 2$ is squarefree and $R \subseteq \{0, 1, \dots, m-1\}$ satisfies $u - v \not\equiv x^3 \pmod m$ for all distinct $u, v \in R$ and all $x \in \mathbb{Z}$; thus $|R| = \alpha_3(m)$ is attainable.

2.1. **Theorem** (Ruzsa)**.** *For every $N \geq 1$ there is $A \subseteq [N]$ with $(A - A) \cap \{x^3 : x \in \mathbb{Z}\} = \{0\}$ and $|A| \geq \frac{|R|}{m^2}\, N^\gamma$, where $\gamma = \frac{1}{3}\big(2 + \log_m |R|\big)$.*

Proof. Let $Y \geq 1$ be an integer and let A_0 be the set of integers $u = \sum_{0\leq i<Y} u_i m^i$ with digits $u_i \in \{0, \dots, m-1\}$ such that $u_i \in R$ whenever $3 \mid i$. Every element of A_0 satisfies $0 \leq u < m^Y$.

Suppose $u, v \in A_0$ are distinct and $u - v = x^3$ for some integer x; then $x \neq 0$. Let j be the least index with $u_j \neq v_j$ and put $d = u_j - v_j$, so $0 < |d| < m$ and

$$u - v = m^j(d + mz) \qquad \text{for some } z \in \mathbb{Z}.$$

Then $m^j \mid u - v$, and $m^{j+1} \nmid u - v$: otherwise $m \mid d + mz$, hence $m \mid d$, which is impossible.

For each prime $p \mid m$ let v_p denote the p-adic valuation, and put $\ell = \min_{p\mid m} v_p(x)$ and $y = x/m^\ell$, an integer since m is squarefree. For every $p \mid m$ we have $v_p(x^3) = 3v_p(x) \geq 3\ell = v_p(m^{3\ell})$, so $m^{3\ell} \mid x^3$; and for a prime $p_0 \mid m$ attaining the minimum, $v_{p_0}(x^3) = 3\ell < 3\ell + 1 = v_{p_0}(m^{3\ell+1})$, so $m^{3\ell+1} \nmid x^3$. Since $x^3 = u - v$, the two facts $m^{3\ell} \mid u - v$, $m^{j+1} \nmid u - v$ give $3\ell \leq j$, and the two facts $m^j \mid x^3$, $m^{3\ell+1} \nmid x^3$ give $j \leq 3\ell$. Hence $j = 3\ell$ is divisible by 3, so $u_j, v_j \in R$. Dividing $u - v = x^3$ by $m^{3\ell}$ gives $d + mz = y^3$, i.e. $u_j - v_j \equiv y^3 \pmod m$ with $u_j \neq v_j$ in R, contradicting the defining property of R. Thus A_0 has no nonzero cube difference.

Now count. The indices $i \in \{0, \dots, Y-1\}$ divisible by 3 number $\lceil Y/3\rceil$, so

$$|A_0| = |R|^{\lceil Y/3\rceil}\, m^{\,Y - \lceil Y/3\rceil}.$$

Writing $\lceil Y/3\rceil = Y/3 + \theta$ with $0 \leq \theta < 1$ and using $|R| \leq m$,

$$|A_0| = |R|^{Y/3} m^{2Y/3} \Big(\frac{|R|}{m}\Big)^\theta \;\geq\; \frac{|R|}{m}\,\big(m^Y\big)^\gamma,$$

because $|R|^{Y/3} m^{2Y/3} = (m^Y)^{(2+\log_m |R|)/3}$.

Given $N \geq m$, let $Y = \lfloor \log_m N\rfloor \geq 1$, so that $m^Y \leq N < m^{Y+1}$, and let $A = A_0 + 1 \subseteq [1, m^Y] \subseteq [N]$. Then $|A| \geq \frac{|R|}{m}(m^Y)^\gamma \geq \frac{|R|}{m}(N/m)^\gamma \geq \frac{|R|}{m^2} N^\gamma$, using $\gamma \leq 1$. For $1 \leq N < m$ the singleton $A = \{1\}$ satisfies $1 \geq \frac{|R|}{m^2} N^\gamma$ since $|R| \leq m$ and $N^\gamma < m$. □

3. The modular problem

For a graph G we write $\alpha(G)$ and $\omega(G)$ for its independence and clique numbers. The strong product $G \boxtimes H$ has vertex set $V(G)\times V(H)$, with $(g, h) \neq (g', h')$ adjacent when ($g = g'$ or $g \sim g'$) and ($h = h'$ or $h \sim h'$). For $m \geq 2$ let $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$ be the graph on $\mathbb{Z}/m\mathbb{Z}$ with $x \sim y$ when $x - y$ is

a nonzero cube modulo m; then $\alpha_3(m) = \alpha(\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z}))$, and we write $\omega_3(m) = \omega(\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z}))$. All these graphs are Cayley graphs of $\mathbb{Z}/m\mathbb{Z}$ and hence vertex-transitive.

3.1. **Lemma** (CRT)**.** *If* $\gcd(m_1, m_2) = 1$ *then* $\mathcal{P}_3(\mathbb{Z}/m_1m_2\mathbb{Z}) \cong \mathcal{P}_3(\mathbb{Z}/m_1\mathbb{Z}) \boxtimes \mathcal{P}_3(\mathbb{Z}/m_2\mathbb{Z})$ *under the Chinese remainder isomorphism* $x \mapsto (x \bmod m_1, x \bmod m_2)$.

Proof. An integer d is a cube modulo m_1m_2 if and only if it is a cube modulo m_1 and modulo m_2 (if $d \equiv x_1^3 \pmod{m_1}$ and $d \equiv x_2^3 \pmod{m_2}$, choose $x \equiv x_i \pmod{m_i}$ by the Chinese remainder theorem; the converse is trivial). Since 0 is a cube, "d is a cube modulo m_i" means "$d \bmod m_i$ is 0 or a nonzero cube", i.e. the residues are equal or adjacent in $\mathcal{P}_3(\mathbb{Z}/m_i\mathbb{Z})$. Finally $d \not\equiv 0 \pmod{m_1m_2}$ means the residue pairs are not both equal. This is the definition of adjacency in the strong product. □

Proof of Theorem 1.2(i). Let $p \mid m$ with $p \not\equiv 1 \pmod 3$. If $p = 3$, then $x \mapsto x^3$ is the identity on $\mathbb{F}_3$; if $p \equiv 2 \pmod 3$, then $\gcd(3, p-1) = 1$ and $x \mapsto x^3$ is a bijection of $\mathbb{F}_p^\times$. In both cases every element of $\mathbb{F}_p$ is a cube, so $\mathcal{P}_3(\mathbb{F}_p)$ is the complete graph K_p. For any graph H, an independent set I of $H \boxtimes K_p$ contains at most one vertex in each fibre $\{h\} \times V(K_p)$ (two vertices of a fibre are adjacent), so its projection to H is injective; the projection is independent in H, since $(h, g) \sim (h', g')$ whenever $h \sim h'$. Hence $\alpha(H \boxtimes K_p) \le \alpha(H)$, while $I' \times \{0\}$ for an independent set I' of H shows $\alpha(H \boxtimes K_p) \ge \alpha(H)$. By Lemma 3.1 with $m_1 = m/p$, $m_2 = p$, we get $\alpha_3(m) = \alpha_3(m/p)$. Since $\log m > \log(m/p) > 0$ and $\log \alpha_3(m/p) \ge 0$, this gives $\log \alpha_3(m)/\log m \le \log \alpha_3(m/p)/\log(m/p)$, with equality only if $\log \alpha_3(m/p) = 0$. □

3.2. **Lemma.** *Let* $p \neq 3$ *be prime. The nonzero cubes modulo* p^3 *are exactly the residues* d *with* $p \nmid d$ *and* $d \bmod p$ *a cube in* $\mathbb{F}_p$*. Consequently* $\mathcal{P}_3(\mathbb{Z}/p^3\mathbb{Z})$ *is obtained from* $\mathcal{P}_3(\mathbb{F}_p)$ *by replacing each vertex by an independent set of size* p^2 *(the fibre of reduction modulo* p*), two vertices being adjacent exactly when their reductions modulo* p *are adjacent.*

Proof. If $p \mid x$ then $p^3 \mid x^3$, so $x^3 \equiv 0$. If $p \nmid x$ then x^3 is a unit whose reduction is a cube in $\mathbb{F}_p$. Conversely, let $p \nmid d$ with $d \equiv c^3 \pmod p$; then $p \nmid c$ and $f(t) = t^3 - d$ satisfies $f(c) \equiv 0$, $f'(c) = 3c^2 \not\equiv 0 \pmod p$ (as $p \neq 3$), so by Hensel's lemma f has a root modulo p^3, i.e. d is a cube modulo p^3. The description of the graph follows: $x \sim y$ in $\mathcal{P}_3(\mathbb{Z}/p^3\mathbb{Z})$ iff $x - y$ is a nonzero cube iff $x \not\equiv y \pmod p$ and $(x - y) \bmod p$ is a nonzero cube in $\mathbb{F}_p$. □

Proof of Theorem 1.2(ii). Let $m = p_1 \cdots p_s$ be squarefree with $3 \nmid m$. By Lemma 3.1 applied to $m^3 = p_1^3 \cdots p_s^3$ and Lemma 3.2, a residue $x \in \mathbb{Z}/m^3\mathbb{Z}$ corresponds to a pair $(a(x), \lambda(x))$ with $a(x) = x \bmod m \in \mathbb{Z}/m\mathbb{Z}$ and a *label* $\lambda(x) = (\lfloor (x \bmod p_i^3)/p_i \rfloor)_i \in \prod_i \mathbb{Z}/p_i^2\mathbb{Z}$ (the map $x \mapsto (a(x), \lambda(x))$ is a bijection), and $x \sim y$ in $\mathcal{P}_3(\mathbb{Z}/m^3\mathbb{Z})$ if and only if $x \neq y$ and, for every i, either $x \equiv y \pmod{p_i^3}$ or $(x - y) \bmod p_i$ is a nonzero cube in $\mathbb{F}_{p_i}$.

Lower bound. Let I be an independent set of $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$ and $R' = \{x : a(x) \in I\}$, so $|R'| = m^2|I|$. Let $x \neq y \in R'$. If $a(x) = a(y)$ then $x \equiv y \pmod m$ but $x \not\equiv y \pmod{m^3}$, so for some i we have $p_i \mid x - y$ and $p_i^3 \nmid x - y$; then $(x - y) \bmod p_i^3$ is neither 0 nor a unit, hence not a cube modulo p_i^3 by Lemma 3.2, and $x \not\sim y$. If $a(x) \neq a(y)$, then $x \sim y$ would force, for each i, that $(x - y) \bmod p_i$ is 0 or a nonzero cube, i.e. $a(x)$ and $a(y)$ equal or adjacent in each factor $\mathcal{P}_3(\mathbb{F}_{p_i})$, so adjacent in $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$ (Lemma 3.1), contradicting the independence of I. Hence R' is independent and $\alpha_3(m^3) \ge m^2\alpha_3(m)$.

Upper bound. Let R' be independent in $\mathcal{P}_3(\mathbb{Z}/m^3\mathbb{Z})$. For each label λ consider the slice $R'_\lambda = \{x \in R' : \lambda(x) = \lambda\}$. Distinct $x, y \in R'_\lambda$ have $a(x) \neq a(y)$ (since $x \mapsto (a(x), \lambda(x))$ is injective), so a is injective on R'_λ. We claim $a(R'_\lambda)$ is independent in $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$. Indeed, let $x \neq y \in R'_\lambda$ and suppose $a(x) \sim a(y)$, i.e. for each i the residues $x \bmod p_i$, $y \bmod p_i$ are equal or adjacent in $\mathcal{P}_3(\mathbb{F}_{p_i})$. If they are equal, then together with $\lambda_i(x) = \lambda_i(y)$ this gives $x \equiv y \pmod{p_i^3}$; if they are adjacent, $(x - y) \bmod p_i$ is a nonzero cube. In either case the i-th condition

for $x \sim y$ holds, so $x \sim y$ in $\mathcal{P}_3(\mathbb{Z}/m^3\mathbb{Z})$, a contradiction. Hence $|R'_\lambda| \le \alpha_3(m)$ for each of the $\prod_i p_i^2 = m^2$ labels, and $|R'| \le m^2\alpha_3(m)$.

Finally, Younis's construction [11, Theorem 1.7] with $F(x) = x^3$, base $M = m^3$ and digit set R' gives the exponent $\log_m |R'|/3$, which by the above equals $\frac{1}{3}(2 + \log_m \alpha_3(m)) = \gamma(m)$ when $|R'| = \alpha_3(m^3)$. □

3.3. **Remark.** The prime 3 behaves differently: the units of $\mathbb{Z}/27\mathbb{Z}$ form a cyclic group of order 18 generated by 2, whose cubes are $\{2^{3j} \bmod 27\} = \{1, 8, 10, 26, 19, 17\}$, i.e. exactly the units $\equiv \pm 1 \pmod 9$, while $27 \mid x^3$ whenever $3 \mid x$. Hence $x \sim y$ in $\mathcal{P}_3(\mathbb{Z}/27\mathbb{Z})$ iff $x - y \equiv \pm 1 \pmod 9$, so $\mathcal{P}_3(\mathbb{Z}/27\mathbb{Z})$ is the 9-cycle C_9 with each vertex replaced by an independent set of size 3, and $\alpha_3(27) = 12 > 9\alpha_3(3) = 9$. The slice argument above applies verbatim (with the 3-adic label being $\lfloor (x \bmod 27)/9 \rfloor$) and gives, for $m = 3m'$ with $3 \nmid m'$ squarefree, $\alpha_3(m^3) = 3(m')^2\,\alpha(C_9 \boxtimes \mathcal{P}_3(\mathbb{Z}/m'\mathbb{Z}))$. By Lemma 3.4 (which applies to any Cayley first factor) and $\omega(C_9) = 2$, this is at most $3(m')^2 \cdot \frac{9}{2}\alpha_3(m') = 13.5\,(m')^2\alpha_3(m')$. The base-$m^3$ construction with this m has exponent $\log_m \alpha_3(m^3)/3$, which exceeds γ_{91} only if $\alpha_3(m^3) > m^{2+r} = 3^{2+r}(m')^{2+r} > 17.1\,(m')^{2+r}$, hence only if $\alpha_3(m') > 1.26\,(m')^r$, i.e. only if m' already beats the record by itself. So the prime 3 never helps either; we do not use this remark elsewhere.

3.4. **Lemma** (Ceiling)**.** *Let G be a Cayley graph of a finite abelian group Γ (with $|\Gamma| = n$) and H any graph. Then $\alpha(G \boxtimes H) \le \frac{n}{\omega(G)}\,\alpha(H)$.*

Proof. Let $K \subseteq \Gamma$ be a clique of G of size $\omega = \omega(G)$. Each translate $K + t$ ($t \in \Gamma$) is again a clique, and each $g \in \Gamma$ lies in exactly ω translates, namely those with $t \in g - K$. Let I be an independent set of $G \boxtimes H$. For a clique C of G put $I_C = \{h \in V(H) : (g, h) \in I \text{ for some } g \in C\}$. For fixed h, the set $\{g : (g, h) \in I\}$ is independent in G (two of its elements $g \sim g'$ would give adjacent vertices $(g, h), (g', h)$), so it meets C in at most one element; hence $|I_C| = |\{(g, h) \in I : g \in C\}|$. Moreover I_C is independent in H: if $h \ne h'$ in I_C were adjacent, with witnesses $g, g' \in C$, then $g = g'$ or $g \sim g'$, so $(g, h) \sim (g', h')$ in $G \boxtimes H$, contradicting independence. Therefore

$$\omega\,|I| = \sum_{t \in \Gamma} \big|\{(g, h) \in I : g \in K + t\}\big| = \sum_{t \in \Gamma} |I_{K+t}| \le n\,\alpha(H). \qquad \square$$

Proof of Theorem 1.2(iii). Apply Lemma 3.4 to $G = \mathcal{P}_3(\mathbb{Z}/m_1\mathbb{Z})$, $H = \mathcal{P}_3(\mathbb{Z}/m_2\mathbb{Z})$, and Lemma 3.1; the floor may be taken since $\alpha_3(m)$ is an integer. For $m_1 = 7$: the cubes modulo 7 are $\{0, 1, 6\}$, so $\mathcal{P}_3(\mathbb{F}_7) = C_7$, $\omega_3(7) = 2$, and the bound is $\lfloor 7\alpha_3(m')/2 \rfloor$. □

3.5. **Remark.** Lemma 3.4 is the classical inequality $\alpha(G \boxtimes H) \le \alpha^*(G)\alpha(H)$ of Hales [5] (α^* the fractional independence number), specialised to vertex-transitive G, where $\alpha^*(G) = n/\omega(G)$; we included the short proof to keep the note self-contained. Note that the ceiling can be attained: $\alpha_3(91) = 14 = \lfloor 7 \cdot 4/2 \rfloor$ and, by the searches of Section 5, $\alpha_3(259) \ge 24 = \lfloor 7 \cdot 7/2 \rfloor$.

4. Exact computations and exclusions

All exact values in this section were obtained with the branch-and-bound clique solver of the `igraph` library, applied to the complement of $\mathcal{P}_3(\mathbb{Z}/m\mathbb{Z})$ (for α_3) or to the graph itself (for ω_3); the reproduction script listed in Appendix A recomputes every entry.

In addition, $\alpha_3(91) = 14$ exactly (the complement of $\mathcal{P}_3(\mathbb{Z}/91\mathbb{Z})$ has clique number 14; this took about half a minute), confirming that the record parameters $(m, |R|) = (91, 14)$ cannot be improved at the modulus 91 itself. A witness is

$$R_{91} = \{7, 9, 12, 18, 29, 35, 38, 40, 55, 58, 60, 78, 80, 83\}.$$

TABLE 1. Exact values for primes $p \equiv 1 \pmod 3$, $p \le 163$. $T(n)$ is the least integer exceeding n^r; "$\lfloor 7\alpha/2 \rfloor$" is the ceiling of Theorem 1.2(iii) for the modulus $7p$.

p	α_3	ω_3	$\log_p \alpha_3$	$T(p)$	$\lfloor 7\alpha_3/2 \rfloor$	$T(7p)$	p	α_3	ω_3	$\log_p \alpha_3$	$T(p)$	$\lfloor 7\alpha_3/2 \rfloor$	$T(7p)$
7	3	2	0.5646	4	—	—	79	9	3	0.5029	13	31	41
13	4	2	0.5405	5	14	15	97	8	5	0.4546	15	28	46
19	4	3	0.4708	6	14	18	103	9	5	0.4741	16	31	47
31	6	3	0.5218	8	21	24	109	9	5	0.4684	16	31	49
37	7	3	0.5389	9	24	26	127	11	3	0.4950	18	38	54
43	7	3	0.5174	10	24	29	139	10	4	0.4666	18	35	56
61	8	3	0.5058	12	28	35	151	11	5	0.4779	19	38	59
67	8	3	0.4946	12	28	37	157	11	5	0.4742	20	38	61
73	8	4	0.4847	13	28	39	163	10	6	0.4520	20	35	62

Proof of Theorem 1.3. (i) By Theorem 1.2(iii), $\alpha_3(7m') \le \frac{7}{2}\alpha_3(m')$. If $\gamma(7m') > \gamma_{91}$ then $\alpha_3(7m') > (7m')^r$, so $\alpha_3(m') > \frac{2}{7}7^r(m')^r = 2 \cdot 7^{r-1}(m')^r$, and $2 \cdot 7^{r-1} = 0.891974\ldots$ (Appendix A checks this constant).

(ii) For $m = 7p$ with $p \le 163$, Theorem 1.2(i) disposes of $p \not\equiv 1 \pmod 3$ (then $\gamma(7p) \le \gamma(7) < \gamma_{91}$, the last inequality because $\alpha_3(7) = 3 < T(7) = 4$), and for $p \equiv 1 \pmod 3$, $p \ne 7$, the last two columns of Table 1 show $\lfloor 7\alpha_3(p)/2 \rfloor < T(7p)$ in every case.

(iii) Apply Theorem 1.2(iii) in both orders to $m = pq$: $\alpha_3(pq) \le \min\big(\lfloor \frac{p}{\omega_3(p)}\alpha_3(q) \rfloor, \lfloor \frac{q}{\omega_3(q)}\alpha_3(p) \rfloor\big)$, and compare with $T(pq)$; the script enumerates the 153 pairs and outputs the 38 survivors of Table 2; that none of them is divisible by 7 is (ii).

(iv) $3367^r = 115.770\ldots$, so $T(3367) = 116$. For the upper bounds use $3367 = 7 \cdot 481$ and $481 = 13 \cdot 37$: $\alpha_3(481) \le \lfloor \frac{13}{2} \cdot 7 \rfloor = 45$ by Theorem 1.2(iii) with $m_1 = 13$, $m_2 = 37$, and then $\alpha_3(3367) \le \lfloor \frac{7}{2} \cdot 45 \rfloor = 157$; if $\alpha_3(481) = 34$ the same step gives $\lfloor \frac{7}{2} \cdot 34 \rfloor = 119$. □

TABLE 2. The 38 two-prime moduli pq, $p < q \le 163$, $p \equiv q \equiv 1 \pmod 3$, not excluded by Theorem 1.2(iii). "need" is $T(pq)$, "ceiling" is the bound of Theorem 1.2(iii), and "found" is the largest independent set produced by the local search of Section 5 (blank when not searched).

m	need	ceiling	found	m	need	ceiling	found	m	need	ceiling	found
403	34	39	27	3139	112	114		9211	209	223	
481	38	45	34	3397	117	129		9313	210	223	
559	41	45	35	4087	130	162		9577	214	223	
793	50	52	36	4453	137	146		10033	220	289	
1027	58	58	42	4819	143	183		10117	221	241	
1147	62	72	45	4891	145	146		10519	226	245	
1333	68	72	45	5293	151	201		10981	232	263	
1591	75	86	57	5461	154	157		11929	243	271	
2257	92	98	57	5767	159	164		12403	249	282	
2479	97	98	61	6649	173	174		17653	306	382	
2623	101	114	59	7747	189	223		19177	321	332	
2881	106	114	63	8479	199	203		19939	328	345	
2923	107	111	66	8509	200	245					

5. Searches

This section reports heuristic lower bounds. The code (a penalty-based local search for maximum independent sets in circulant graphs, and simulated-annealing and tabu searches over the fibre decomposition described below) accompanies the paper; every value below is reproducible in minutes, and every value was verified to be an independent set. We make no claim that any of these values is optimal.

5.1. **Two-prime moduli.** For every squarefree $m \le 3000$ composed of two primes $\equiv 1 \pmod 3$, a local search of four to six seconds per modulus found the values in Table 3. None reaches $T(m)$, and none comes within 2% of the record ratio except $m = 91$ itself and $m = 259$, 481 (ratios 0.5719, 0.5710). For the moduli $7p$ this is consistent with (and implied by) Theorem 1.3(ii); for the others it is search evidence only. Longer runs (twenty seconds) gave $\alpha_3(1147) \ge 46$ and $\alpha_3(1591) \ge 58$.

Table 3. Search lower bounds for all squarefree $m \le 3000$ with exactly two prime factors, both $\equiv 1 \pmod 3$ (four to six seconds of local search each). The ratio is $\log_m(\text{found})$; the record ratio is $r = 0.58505$. Moduli $7p$ are included for completeness although they are excluded by Theorem 1.3(ii).

m	need	found	ratio	m	need	found	ratio	m	need	found	ratio
91	15	14	0.5850	1147	62	45	0.5403	2119	89	43	0.4911
217	24	19	0.5473	1159	63	37	0.5118	2257	92	57	0.5236
247	26	21	0.5526	1261	66	37	0.5058	2263	92	48	0.5012
259	26	24	0.5719	1273	66	40	0.5160	2353	94	47	0.4959
301	29	24	0.5569	1333	68	45	0.5291	2413	96	49	0.4997
403	34	27	0.5494	1339	68	41	0.5158	2449	97	56	0.5158
427	35	26	0.5379	1387	69	38	0.5028	2479	97	61	0.5260
469	37	28	0.5418	1417	70	45	0.5246	2509	98	55	0.5119
481	38	34	0.5710	1501	73	43	0.5143	2587	100	54	0.5076
511	39	26	0.5224	1591	75	57	0.5484	2623	101	59	0.5180
553	41	31	0.5438	1651	77	47	0.5196	2641	101	50	0.4965
559	41	35	0.5620	1807	81	48	0.5162	2701	102	57	0.5117
703	47	37	0.5508	1843	82	37	0.4802	2743	103	48	0.4890
763	49	31	0.5174	1891	83	49	0.5158	2869	106	45	0.4781
793	50	36	0.5368	1957	85	41	0.4900	2881	106	63	0.5201
817	51	36	0.5344	1963	85	45	0.5021	2899	107	47	0.4830
871	53	40	0.5449	2041	87	49	0.5107	2923	107	66	0.5250
949	56	38	0.5306	2071	88	48	0.5070	2977	108	48	0.4840
1027	58	42	0.5390	2077	88	52	0.5173	2983	108	51	0.4914

5.2. **Three-prime moduli.** For $m = 7\cdot13\cdot p$ the ceiling $\lfloor \frac{7}{2}\alpha_3(13p) \rfloor$ combined with the searched values of $\alpha_3(13p)$ leaves, among $p \in \{19, 31, 37, 43, 61\}$, only $p = 37$: for $p = 19$ the ceiling is at most $\lfloor 3.5\cdot 21 \rfloor = 73 < 79 = T(1729)$ if $\alpha_3(247) = 21$; for $p = 31$, $\lfloor 3.5\cdot 27 \rfloor = 94 < 105 = T(2821)$; for $p = 43$, $\lfloor 3.5\cdot 35 \rfloor = 122 < 127 = T(3913)$; for $p = 61$, $\lfloor 3.5\cdot 36 \rfloor = 126 < 156 = T(5551)$ (these exclusions are conditional on the searched values of $\alpha_3(13p)$ being exact). Direct searches gave $\alpha_3(1729) \ge 67$, $\alpha_3(2821) \ge 87$, and, for $m = 3367$, the plain local search on the 3367-vertex graph gave only 104–106; the structured search below did much better.

5.3. **The modulus 3367.** Since $\mathcal{P}_3(\mathbb{F}_7) = C_7$, Lemma 3.1 gives $\mathcal{P}_3(\mathbb{Z}/3367\mathbb{Z}) \cong C_7 \boxtimes H$ with $H = \mathcal{P}_3(\mathbb{Z}/481\mathbb{Z})$, a 481-vertex graph of degree 64. An independent set I of $C_7 \boxtimes H$ is the same as an assignment $h \mapsto F_h \subseteq \mathbb{Z}/7\mathbb{Z}$ ($h \in \mathbb{Z}/481\mathbb{Z}$) of *fibres* $F_h = \{c : (c,h) \in I\}$ such that each F_h is independent in C_7 and, whenever $h \sim h'$ in H, the sets F_h and $F_{h'}$ are disjoint and contain no two adjacent vertices of C_7; then $|I| = \sum_h |F_h|$. There are 29 independent subsets of C_7 (the empty set, 7 singletons, 14 pairs and 7 triples), so the search space consists of 29^{481} fibre assignments. Equivalently, writing $J_c = \{h : c \in F_h\}$, the seven "consecutive unions" $K_c = J_c \cup J_{c+1}$ must be independent sets of H, with $J_c \cap J_{c+1} = \emptyset$, and $|I| = \frac{1}{2}\sum_c |K_c|$; this is how the ceiling $\alpha(C_7 \boxtimes H) \le \frac{7}{2}\alpha(H)$ of Lemma 3.4 manifests itself in this case.

We ran two searches over fibre assignments, with the objective $\sum_h |F_h|$ minus an adaptive penalty for each violated pair: a simulated annealing (about 10^9 single-fibre moves per run) and a tabu search (best admissible single-fibre move at each step, incremental score tables, tabu tenure 8 to 12 steps). Over about fifteen annealing runs the best values were 115 (twice), 114, 113 and 111; the tabu search reached 116 in one of five cold-started runs of 40 seconds (the others ended between 99 and 102), at which point it was stopped; runs warm-started from the 115- and 116-element sets returned 115 and 116 and did not find 117. The winning assignment uses 50 singleton fibres and 33 pairs; its seven consecutive unions have sizes $32, 34, 34, 34, 30, 34, 34$, five of them attaining the searched value $\alpha_3(481) \ge 34$. A 117-element set would require the seven unions to have total size 234.

6. The certificate

The set $R_{3367} \subseteq \mathbb{Z}/3367\mathbb{Z}$ below has 116 elements and no two of them differ by a nonzero cube modulo 3367; verification is the finite computation performed by the script of Appendix A. Its elements reduce modulo 7 to c and modulo 481 to h exactly when $c \in F_h$ in the fibre description of Section 5.3.

$$\begin{aligned} R_{3367} = \{&3, 10, 27, 101, 119, 130, 210, 214, 221, 291, 295, 312, 343, 365, 375,\\ &382, 386, 434, 456, 526, 617, 628, 698, 702, 709, 719, 772, 789, 793, 800,\\ &863, 880, 891, 1007, 1018, 1035, 1070, 1098, 1105, 1109, 1126, 1161, 1179, 1189, 1196,\\ &1200, 1229, 1281, 1320, 1372, 1390, 1442, 1464, 1481, 1512, 1533, 1555, 1603, 1625, 1688,\\ &1716, 1779, 1797, 1871, 1888, 1895, 1906, 1962, 1979, 1986, 1990, 2106, 2117, 2169, 2180,\\ &2197, 2204, 2208, 2225, 2260, 2271, 2278, 2295, 2369, 2387, 2391, 2471, 2482, 2489, 2563,\\ &2580, 2611, 2654, 2661, 2702, 2706, 2752, 2794, 2878, 2896, 2970, 2987, 3005, 3040, 3057,\\ &3061, 3096, 3148, 3216, 3275, 3279, 3286, 3303, 3307, 3342, 3359\}. \end{aligned}$$

Proof of Theorem 1.1. The nonzero cubes modulo 3367 number $3 \cdot 5 \cdot 13 - 1 = 194$ (there are $1 + (p-1)/3$ cubes in $\mathbb{F}_p$ for $p \equiv 1 \pmod 3$, and Lemma 3.1); the script checks that none of the $116 \cdot 115$ ordered differences of elements of R_{3367} is among them. Thus $\alpha_3(3367) \ge 116$. Theorem 2.1 with $m = 3367$, $R = R_{3367}$ gives $D_3(N) \ge \frac{116}{3367^2} N^\gamma$ with $\gamma = \frac{1}{3}(2 + \log_{3367} 116) = 0.861763131630587\ldots$ (computed with 20 significant digits), and $\gamma - \gamma_{91} = 8.135 \cdot 10^{-5}$. □

7. Remarks and open questions

(1) The window for the modulus 3367 is $116 \le \alpha_3(3367) \le 157$ unconditionally and ≤ 119 if $\alpha_3(481) = 34$; each additional element modulo 3367 would raise the exponent by about $8 \cdot 10^{-5}$. An exact computation of $\alpha_3(481)$ (a 481-vertex graph of degree 64) is within reach of a dedicated maximum-clique code with symmetry breaking and would sharpen the window; our attempts with a general mixed-integer solver did not close the gap.

(2) By Theorem 1.3(i), any further improving modulus divisible by 7 has cofactor m' with $\alpha_3(m') > 0.892\,(m')^r$. Among the cofactors we searched, 481 is the only one with this property; the next three-prime moduli $7\cdot 13\cdot p$ for $p \in \{19, 31, 43, 61\}$ are excluded by the ceiling if the searched values $\alpha_3(13p) \in \{21, 27, 35, 36\}$ are exact. Moduli not divisible by 7 start from the weaker single-prime ratios of Table 1 (≤ 0.5405) and would need a strong-product gain of more than 8% in the exponent, which none of the searched products approached (Table 3). We therefore expect 3367, or a larger multiple of 91 with a cofactor as good as 481, to be the only places to look.

(3) Theorem 1.2(ii) and Remark 3.3 show that the two natural digital variants of Ruzsa's construction coincide for cubes. Younis's Proposition 5.1 [11] proves $r_k(p^k) = p^{k-1} r_k(p)$ for primes $p \nmid k$; the same slice argument, with 3 replaced by k and $p \nmid k$ in Lemma 3.2, gives $r_k(m^k) = m^{k-1} r_k(m)$ for squarefree m coprime to k.

(4) The fibre formulation of Section 5.3 applies to squares as well, with $\mathcal{P}_2(\mathbb{F}_5) = C_5$: an improvement of the exponent $\frac{1}{2}(1 + \log_{205} 12)$ at a modulus $5m'$ requires $\alpha_2(5m') > (5m')^{\log_{205} 12}$ with the ceiling $\lfloor 5\alpha_2(m')/2 \rfloor$. Our tabu search reached 37 at $m = 2665 = 5 \cdot 13 \cdot 41$ (need 40, ceiling 42 if $\alpha_2(533) = 17$) and 30 at $m = 1885$ (need 34); we leave this as an open computational problem.

Appendix A. Verification script

The following Python script uses only the standard library and checks every finite assertion of this note: the record witness R_{91}, the certificate R_{3367}, the constants r, γ_{91}, $\gamma(3367)$, the inequality $\gamma(3367) > \gamma_{91}$, $T(3367) = 116$, and the clique numbers $\omega_3(7) = \omega_3(13) = 2$, $\omega_3(37) = 3$ used in Theorem 1.3(iv). A second script, `reproduce.py` (requiring `python-igraph`), recomputes Table 1, $\alpha_3(91) = 14$, and the exclusion lists of Theorem 1.3; the search codes `mis.c`, `fiber.c` and `tabu.c` reproduce Section 5 (the certificate is the output of `tabu` with seed 3 and tenure 8).[1]

```
#!/usr/bin/env python3
"""Standalone verification (Python standard library only) of the finite claims in
"An improved lower bound for sets with no cube differences".

Checks:
 (1) the 14-element set R91 in Z/91 has no nonzero cube difference (Lewko/Ruzsa record parameters);
 (2) the 116-element set R3367 in Z/3367 has no nonzero cube difference;
 (3) the exponent arithmetic: gamma(91,14) < gamma(3367,116), and that 116 is the threshold at 3367;
 (4) the clique numbers omega(Paley_3(F_7)) = 2, omega(Paley_3(F_13)) = 2, omega(Paley_3(F_37)) = 3
     (brute force), which enter the Hales ceiling 3.5*alpha(481) for m = 3367.
"""
import json, math, itertools, os, sys

R91 = [7, 12, 58, 83, 35, 29, 9, 55, 60, 38, 18, 78, 80, 40]

HERE = os.path.dirname(os.path.abspath(__file__))
cert = json.load(open(os.path.join(HERE, "certificate.json")))
R3367 = cert["R"]


def cubes(m):
    return {pow(x, 3, m) for x in range(m)} - {0}
```

[1] Code and data accompany the paper as arXiv ancillary files.

```
def cube_free(R, m):
    C = cubes(m)
    return all((a - b) % m not in C for a, b in itertools.permutations(R, 2))


def omega(p):
    C = cubes(p) | {(-c) % p for c in cubes(p)}
    best = 1
    for k in range(2, 8):
        found = False
        for S in itertools.combinations(range(1, p), k - 1):
            T = (0,) + S
            if all((a - b) % p in C for a, b in itertools.combinations(T, 2)):
                found = True
                break
        if not found:
            return best
        best = k
    return best


assert len(set(R91)) == 14 and cube_free(R91, 91)
assert len(set(R3367)) == 116 and cube_free(R3367, 3367)
r91 = math.log(14) / math.log(91)
assert abs(r91 - 0.58504533909837) < 1e-12
assert abs((2 + r91) / 3 - 0.86168177969946) < 1e-12
assert math.floor(3367 ** r91) + 1 == 116 and 115 < 3367 ** r91 < 116
r3367 = math.log(116) / math.log(3367)
assert r3367 > r91
assert abs((2 + r3367) / 3 - 0.86176313163059) < 1e-12
assert omega(7) == 2 and omega(13) == 2 and omega(37) == 3
print("all asserts passed")
print("ratio log14/log91   = %.15f" % r91)
print("gamma_91            = %.15f" % ((2 + r91) / 3))
print("ratio log116/log3367 = %.15f" % r3367)
print("gamma_3367           = %.15f" % ((2 + r3367) / 3))
print("threshold at m=3367: |R| >= %d" % (math.floor(3367 ** r91) + 1))
```

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